

Answer to all questions.

Basic

1. Create an array containing five different fruits.
2. Create an array of numbers from 1 to 10.
3. Find the first element of an array.
4. Find the last element of an array.
5. Add a new item to the end of an array.
6. Add a new item to the beginning of an array.
7. Remove the last item from an array.
8. Remove the first item from an array.
9. Find the total number of elements in an array.
10. Print every element in an array using a for loop.
11. Print every element using a for...of loop.
12. Check whether an array contains a specific value.
13. Find the index of a given value in an array.
14. Replace the third element of an array with a new value.
15. Create a copy of an array.
16. Join all array elements into a single string separated by commas.
17. Reverse an array.
18. Sort an array of numbers in ascending order.
19. Sort an array of strings alphabetically.
20. Create a new array containing only even numbers.
21. Create a new array containing only odd numbers.
22. Find the largest number in an array.
23. Find the smallest number in an array.
24. Calculate the sum of all numbers in an array.
25. Calculate the average of all numbers in an array.

Intermediate Level:

1. Remove duplicate values from an array.
2. Count how many times a specific value appears in an array.
3. Find the second largest number in an array.
4. Find the second smallest number in an array.
5. Merge two arrays into one.
6. Find the common elements between two arrays.
7. Find the elements that exist in the first array but not the second.
8. Rotate an array one position to the right.
9. Rotate an array one position to the left.
10. Reverse an array without using `.reverse()`.
11. Check whether an array is sorted.
12. Split an array into chunks of size 3.

13. Remove all negative numbers from an array.
14. Multiply every element by 2.
15. Convert an array of names into uppercase.
16. Count how many even and odd numbers are in an array.
17. Find the longest string in an array.
18. Find the shortest string in an array.
19. Flatten a one-level nested array.
20. Create an array of squares for every number.
21. Find all numbers greater than 50.
22. Find the first number greater than 100.
23. Check whether every number is positive.
24. Check whether at least one number is negative.
25. Create a frequency object that counts how many times each value appears.

Matrix Operations:

1. Add two matrices.
2. Subtract two matrices.
3. Multiply every element of a matrix by a scalar.
4. Find the transpose of a matrix.
5. Check whether two matrices are equal.
6. Check whether a matrix is square.
7. Check whether a matrix is an identity matrix.
8. Check whether a matrix is a zero matrix.
9. Find the trace of a square matrix.
10. Find the sum of each row.
11. Find the sum of each column.
12. Find the row with the largest sum.
13. Find the column with the largest sum.
14. Swap the first and last rows.
15. Swap the first and last columns.
16. Print the matrix in spiral order.
17. Mirror a matrix horizontally.
18. Mirror a matrix vertically.
19. Find the maximum value in each row.
20. Find the minimum value in each column.
21. Replace all diagonal elements with 0.
22. Count how many elements are greater than the average value.